Palestine Technical University – Kadoorie

Faculty of Information Technology and Artificial Intelligence

Department of Artificial Intelligence | Tulkarm Branch

Ardi

أرضي

AI-Powered Real Estate Monitoring and Fraud Detection System

A Graduation Project Submitted in Partial Fulfillment of the

Requirements for the Bachelor's Degree in Artificial Intelligence

Prepared By	Student ID
Kinan Awartani	202311151
Taima Abu Bshara	202310481
Lujain Manasreh	202310631
Zaid Rajab	202310471
Amer Riad	202220062

Supervised By

Dr. Areej Jaber

Department of Artificial Intelligence

Academic Year 2026 / 2027 — first Semester

Declaration of Originality

We hereby declare that this graduation project, titled “Ardi – AI-Powered Real Estate Monitoring and Fraud Detection System ,” is our own original work carried out under the supervision of Dr. Areej Jaber at Palestine Technical University – Kadoorie.

We further declare that:

1. This work has not been submitted for any other degree at this or any other institution.
2. All sources and references used have been properly cited and acknowledged.
3. Any assistance received from others has been duly acknowledged.
4. The use of generative AI tools, if any, has been disclosed and cited according to the department’s academic integrity policy.

Student Name	Signature	Date
Kinan Awartani	KINAN	7 october 2026
Taima Abu Bshara	TAIMA	7 october 2026
Lujain Manasreh	LUJAIN	7 october 2026
Zaid Rajab	ZAID	7 october 2026
Amer Riad	AMER	7 october 2026

Abstract

Title: Ardi – AI-Powered Real Estate Monitoring and Fraud Detection System

Keywords: Artificial Intelligence, Real Estate, Fraud Detection, Anomaly Detection, Machine Learning, Data Analytics

Ardi is an AI-powered real estate monitoring and fraud detection system designed to support authorities in monitoring and analysing property and land transactions.

The system centralises real estate information including properties, buyers, sellers, transaction values, geographical locations, historical records, and transaction dates.

Using data analytics and machine learning techniques, the system analyses historical and current transaction behaviour to detect unusual patterns and anomalies that may require further investigation.

Rather than automatically classifying individuals or transactions as fraudulent, Ardi operates as a decision-support system that assigns an anomaly or suspicion score to unusual activities and presents the results to authorised users for human review.

The platform also provides an interactive dashboard, geographical visualisation of real estate activity, and periodic reports to assist authorities in understanding market behaviour and identifying transactions that deserve further attention.

الملخص

عنوان المشروع :

أرضي هو نظام ذكي لمراقبة العقارات واكتشاف الأنماط المحتملة للاحتيال

الكلمات المفتاحية:

الذكاء الاصطناعي، العقارات، كشف الاحتيال، اكتشاف الشذوذ، التعلم الآلي، تحليل البيانات

مشروع أرضي هو نظام ذكي قائم على تقنيات الذكاء الاصطناعي يهدف إلى مساعدة الجهات المختصة في مراقبة وتحليل معاملات الأراضي والعقارات.

يقوم النظام بجمع وتنظيم بيانات العقارات والمعاملات والأطراف المرتبطة بها والأسعار والتواريخ والمواقع الجغرافية ضمن منصة مركزية، ثم يستخدم تقنيات تحليل البيانات والتعلم الآلي لتحليل النشاط العقاري الحالي والتاريخي واكتشاف الأنماط والسلوكيات غير الطبيعية التي قد تستحق المزيد من المراجعة والتحقيق.

لا يقوم النظام بإصدار حكم قطعي بأن معاملة معينة احتيالية، بل يعمل كنظام لدعم القرار من خلال تحديد الحالات غير الطبيعية وإعطائها درجة اشتباه أو درجة شذوذ تساعد الموظف المختص على ترتيب أولويات المراجعة.

كما يوفر النظام لوحة تحكم تفاعلية، وتحليلاً جغرافياً للنشاط العقاري، وتقارير دورية تهدف إلى تحويل البيانات الخام إلى معلومات ومؤشرات تساعد الجهات المختصة في عملية المراقبة واتخاذ القرار.

Dedication

To our families, whose endless support and patience made this work possible.

To our supervisor and professors at Palestine Technical University – Kadoorie who guided us throughout this journey.

And to Palestine.

Acknowledgements

We would like to express our sincere gratitude to Dr. Areej Jaber for her invaluable guidance, continuous support, constructive feedback, and encouragement throughout the development of this graduation project.

We are deeply grateful to the Faculty of Information Technology and Artificial Intelligence at Palestine Technical University – Kadoorie for providing the academic environment and resources necessary to complete this work.

We also extend our sincere appreciation to all instructors, colleagues, friends, and individuals who contributed directly or indirectly to this project.

Finally, we would like to thank our families for their continuous support, patience, and encouragement throughout our academic journey.

Contents

Declaration of Originality i

Abstract ii

Arabic Abstract iii

Dedication iv

Acknowledgements v

List of Figures vii

List of Tables viii

1 Introduction 1

1.1 Background and Motivation 1
1.2 Problem Statement 1
1.3 Project Objectives 2
1.4 Research Questions 2
1.5 Significance of the Study 2
1.6 Scope and Limitations 3
1.7 Report Organisation 3

List of Figures

List of Tables

Chapter 1: Introduction

This chapter introduces the research problem, provides background context, and outlines the objectives, research questions, significance, scope, limitations, and overall structure of the report.

1.1 Background and Motivation

Real estate and land transactions generate large amounts of data related to properties, buyers, sellers, transaction values, geographical locations, and historical ownership records.

As the number of transactions increases, manually reviewing and monitoring this information becomes more difficult, especially when data is distributed across multiple sources and systems.

Artificial intelligence and data analytics provide opportunities to analyse historical and current real estate activity automatically, identify recurring patterns, and highlight unusual behaviour that may require additional investigation.

The Ardi project is motivated by the need for a centralised intelligent platform that supports authorities in monitoring property transactions more efficiently and converting large amounts of raw transaction data into useful information.

1.2 Problem Statement

The increasing volume and complexity of real estate transactions make manual monitoring time-consuming and inefficient.

A transaction may appear normal when examined independently, while a historical, relational, or geographical analysis may reveal unusual behaviour.

Examples include significant changes in property prices, frequent transactions between the same parties, repeated resale of the same property within short periods, unusually high transaction activity by a specific person or organisation, or significant differences between the prices of similar properties.

Identifying such patterns manually requires substantial effort and may be difficult when large volumes of data are involved.

Therefore, there is a need for a decision-support system that can centralise real estate data, analyse transaction behaviour, identify unusual patterns, and present suspicious cases

to authorised users for further review.

1.3 Project Objectives

The main objectives of the Ardi project are:

1. To develop a centralised platform for managing property and real estate transaction data.
2. To analyse historical and current real estate activity.
3. To apply artificial intelligence and machine learning techniques for anomaly detection.
4. To identify transactions and behavioural patterns that may require further review.
5. To analyse real estate activity geographically.
6. To provide an interactive dashboard containing useful statistics and key performance indicators.
7. To generate daily, weekly, monthly, and yearly reports.
8. To reduce the time and effort required for manual transaction monitoring.
9. To transform raw real estate data into useful information that supports decision-making.

1.4 Research Questions

This project aims to answer the following research questions:

1. How can artificial intelligence techniques be used to detect unusual patterns in real estate transactions?
2. Which transaction, property, temporal, geographical, and behavioural features are most useful for identifying anomalous real estate activities?
3. How can anomaly detection results be presented in a way that supports human review and decision-making?
4. How can geographical and historical transaction analysis improve the monitoring of real estate activities?

1.5 Significance of the Study

The significance of this project lies in integrating real estate data, artificial intelligence, data analytics, geographical analysis, interactive visualisation, and automated reporting within one

decision-support platform.

Instead of relying entirely on manual inspection, authorised users can use Ardi to identify transactions and activities that deserve greater attention.

The system can also provide a clearer understanding of market trends, transaction activity, unusual behaviour, and geographical concentrations.

1.6 Scope and Limitations

The scope of Ardi includes property and transaction data management, historical transaction analysis, anomaly detection, dashboard visualisation, geographical analysis, and periodic reporting.

The system is intended to support authorised authorities and specialised employees rather than general public users.

The system does not automatically determine that a person or transaction is fraudulent.

It does not replace human investigation, legal procedures, or official decision-making.

The quality and accuracy of the system depend on the availability, quality, completeness, and accuracy of the provided data.

The performance of the machine learning model may also be affected by limited, imbalanced, or incomplete training data.

1.7 Report Organisation

The remainder of this report is organised as follows:

- Chapter 2 presents the theoretical background, related work, and research gap.
- Chapter 3 describes the methodology, system requirements, architecture, database design, data processing, and AI approach.
- Chapter 4 presents the system implementation and experimental results.
- Chapter 5 discusses the results, strengths, limitations, and challenges.
- Chapter 6 concludes the project and presents possible future improvements.